Formula sheet for Quiz 8

Shapes

A batch of token ids is (N, T): N sequences, T positions.

$$\texttt{nn.Embedding}(V, d) \;:\; (N, T) \longrightarrow (N, T, d) \qquad \text{parameters} = V \times d$$

Reserved ids, by convention: `<pad>` $= 0$, `<unk>` $= 1$, real tokens from 2.

Padding

For a batch whose sequences have lengths $\ell_1 \dots \ell_N$:

$$\text{cells} = N \times T \qquad \text{real} = \sum_i \ell_i \qquad \text{padding share} = 1 - \frac{\sum_i \ell_i}{N \times T}$$

Masking

The mask is `ids` $\neq$ `PAD`, shape (N, T). Masked mean:

$$\text{pooled}_i = \frac{\sum_t m_{i,t}\, E_{i,t}}{\sum_t m_{i,t}} \qquad \text{against the unmasked} \quad \frac{1}{T}\sum_t E_{i,t}$$

When the pad id embeds to the zero vector, the unmasked pooled vector is the masked one scaled by ℓ/T, so

$$\frac{\|\text{unmasked} - \text{masked}\|}{\|\text{masked}\|} = 1 - \frac{\ell}{T}$$

For a per-token loss, `F.cross_entropy(..., ignore_index=0)`.

1D convolution

A kernel of width k widens the receptive field by $k-1$ per layer, so L stacked layers see

$$L\,(k-1) + 1 \quad \text{positions} \qquad (k = 3 \Rightarrow 2L + 1)$$

`nn.Conv1d` takes (N, C, T), not (N, T, C).

Recurrence

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b)\,, \qquad h_0 = \mathbf{0}$$

`nn.RNN` and `nn.LSTM` return `(output, h_n)`; with `batch_first=True`, `output` is (N, T, H), the hidden state at every step.

Backpropagation through time

$$\frac{\partial h_T}{\partial h_t} = \prod_{k=t+1}^{T} \frac{\partial h_k}{\partial h_{k-1}}$$

A product of $T - t$ terms: if each is r, the gradient reaching back n steps goes as r^n.

LSTM, and what a sequence costs

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \qquad h_t = o_t \odot \tanh(c_t)$$

Each gate is a learned function of x_t and h_{t-1} squashed to $[0, 1]$. A recurrent layer takes T sequential steps and total work growing with T; self-attention takes 1 sequential step and work growing with T^2.